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PROFILE

AI/ML Engineer, currently an Information Systems undergraduate at Telkom University. Passionate about building intelligent, real-world applications, from training and deploying machine learning models to developing the full-stack web platforms that bring them to life. Holds 5 professional certifications from Google, IBM, Microsoft, and HackerRank, with a proven track record of shipping 6+ end-to-end products and reaching national finalist stages in 3 hackathons within the past 6 months.

EDUCATION

Telkom University | Bachelor of Information Systems

August 2023 – Present

- Current GPA: 3.69/4.00
- Relevant Coursework: Web Application Development, Object-Oriented Programming, Algorithm and Programming, Data Mining, Cloud Computing, Cyber Security, and Enterprise Resource Planning.
- Involvements: Active Member of GDGoC Machine Learning Study Group across 12+ weekly sessions (exploring Python, TensorFlow, and computer vision implementation).

TECHNICAL SKILLS & COMPETENCIES

- Full-Stack Web Development: React, Next.js, TypeScript, Node.js, NestJS, Express.js, FastAPI, PHP, Laravel, PostgreSQL, Prisma, Supabase, RESTful APIs, Docker, Git/GitHub. Integrations: Midtrans, Google OAuth, SMTP, dompdf.
- Artificial Intelligence & ML: Python, TensorFlow / TensorFlow.js, PyTorch, Scikit-Learn, OpenCV, MediaPipe, YOLO. CNNs (2D/1D), transfer learning, model explainability (Grad-CAM), leakage-free evaluation (patient-level splits, ROC/AUC, F1).
- LLM Application Development: OpenAI API, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Regular Expressions.
- Enterprise Systems: Odoo & SAP.
- 3D & Graphics: Three.js, react-three-fiber, on-device inference, real-time computer vision (analytic inverse kinematics).
- Data Science & Analytics: Pandas, NumPy, SQL (Advanced), Matplotlib, Jupyter Notebook, Kaggle, Google Cloud Platform.

WORK EXPERIENCES

Artificial Intelligence Engineer | PT Data Labs Analytics (Internship)

June 2026 – Present

- Delivering AI/ML solutions for an enterprise client as part of the company's outsourced engineering team, contributing across the full model development lifecycle.
- Operating within a client-facing engagement under confidentiality constraints, translating business requirements into technical solutions while following the team's production standards for reproducibility, code quality, and reliable delivery.
- Gained hands-on experience working within the Google Cloud Platform environment, using BigQuery for large-scale data querying and processing as part of the client's cloud-based data workflow.

Machine Learning Engineer | HUMIC Engineering Research Center (Internship)

April 2026 – Present

- Conducted a Systematic Literature Review across IEEE, Scopus, and Google Scholar, appraising 10+ papers and exposing that many 98 to 99% accuracy claims come from patient-level data leakage, which set an honest 0.85 AUC target.
- Built leakage-free PyTorch pipelines on CBIS-DDSM (whole-image and ROI, patient-level split) and trained 6 architectures, from EfficientNet-B3 and ConvNeXt to a CBAM hybrid-fusion network and Mammo-CLIP, across 12+ controlled experiments.
- Raised AUC by ~6 points with clinical-metadata fusion (0.80 to 0.86), evaluated with F1, ROC/AUC, and confusion matrices plus Grad-CAM, and first-authored a 6-page IEEE conference paper.

ORGANIZATIONAL EXPERIENCES

Full-Stack Developer | Central Computer Improvement (CCI) Telkom University

February 2026 – Present

- Served as backend developer on a 6-person engineering team building a vendor marketplace platform in partnership with UKM CSC, developing 90+ RESTful API endpoints using NestJS, PostgreSQL, and Prisma ORM. Authored full API documentation, tested endpoints with Postman, and configured Supabase, Midtrans payment gateway, SMTP, and Google OAuth integrations.
- Independently built a full stack document automation platform for Telkom University's student directorate, enabling 50+ student organizations to auto generate Proposal Kegiatan and LPJ documents, cutting processing time from 2+ hours to minutes. Designed the system from scratch with no predefined spec, replicated official templates via dompdf, and passed a penetration test from Telkom's PuTI Directorate before deployment.

Laboratory Assistant | ERP Laboratory Telkom University

March 2026 – Present

- Lead the Champions Teams work program, an initiative aimed at increasing competitive participation among 200+ Information Systems students through curated competition opportunities.
- Research, curate, and disseminate 10+ national and international competitions every month across Essay, Business Model Canvas, Design, IT, and Business Plan categories.

Sponsorship Staff | Telkom University National Futsal Championship

December 2025 – May 2026

- Identified and approached 40+ potential corporate sponsors through targeted B2B outreach, conducting 15+ pitching sessions to companies and negotiating partnership agreements covering branding placements and event activation.
- Established partnership agreements with multiple tenants and managed tenant coordination throughout the event, contributing to a 90% sponsorship slot fulfilment rate ahead of competition day.

PROJECTS**KawanTuli | Capstone Project**

July 2026 - Present

- Developing KawanTuli, a BISINDO (Indonesian Sign Language) accessibility platform combining a two-way translator with an avatar-based learning system, built with Next.js, TypeScript, and TensorFlow.js, with all camera inference running on-device so no video ever leaves the browser.
- Trained and deployed two recognition models: a 26-class alphabet classifier at 99.31% validation accuracy (~9,700 images), and a 32-class word-level 1D CNN at 83.13% signer-independent accuracy (1,600 videos), improved from 75.31% by fusing MediaPipe hand landmarks with arm pose features, running in real time at 15 fps with 6 ms median latency.
- Built a rigged 3D avatar (react-three-fiber) that performs sign poses derived from dataset landmarks via analytic IK, plus two internal authoring tools enabling non-engineers to correct poses and re-record gestures via webcam.

Mammography Benign vs Malignant Research Paper | HUMIC Engineering Research Center

April 2026 – July 2026

- Within a 3-person team where each member independently trained and fine-tuned their own models, my leakage-free approach produced the best results and was selected as the team's final pipeline.
- Built the winning solution end-to-end: a fair whole-image versus ROI comparison on CBIS-DDSM with patient-level splitting, benchmarking 6 architectures including a custom CBAM hybrid-fusion model and Mammo-CLIP across 12+ experiments.
- Pinpointed clinical metadata as the biggest lever (AUC 0.80 to 0.86) with Grad-CAM evidence of correct lesion focus and authored the Methodology and Results and Discussion chapters of the resulting 6-page IEEE paper.

AMANIN | Social Project

April 2026 – May 2026

- Developed AMANIN, an AI-powered image and video detection app designed to prevent media-based fraud by leveraging self-built deepfake detection models and digital forensic techniques, building the entire web application independently from end to end.
- Trained a custom deepfake detection model from scratch using PyTorch and OpenCV on a dataset of 10,000+ images and videos, achieving 90%+ detection accuracy in classifying AI-generated and manipulated media.
- Delivered as a social impact initiative educating 40+ residents across 1 rural village, raising community awareness of the dangers of fraud using AI-generated images and videos and equipping them to recognize manipulated media.

MLCRAFT | Open-Source AI/ML Engineering Plugin

July 2026

- Built mlcrafter, an open-source plugin that turns AI coding assistants (Claude Code, Antigravity) into a senior AI/ML research engineer, packaging 23 skills that auto-route a plain-language task to the best method across 14 domains including computer vision, medical imaging, NLP and LLM, tabular, time-series, generative, reinforcement learning, graph, multimodal, video, and 3D.
- Encoded a research-first, leakage-aware workflow grounded in 2021 to 2026 literature: paper review with realistic-ceiling estimation, group, patient, and temporal split audits, per-problem metric and explainability selection (ROC-AUC vs mAP vs RMSE, Grad-CAM vs bounding boxes vs SHAP), and an iterative accuracy-improvement loop.

GAP | Personal Project

May 2026 – June

- Developing GAP, an API intelligence platform built with TypeScript that auto-detects routes, request bodies, and backend-to-frontend contract mismatches directly from source code, leveraging Retrieval-Augmented Generation (RAG) and Regular Expressions, with zero annotations and zero manual documentation required.
- Designed the core engine that parses entire codebases and generates a full API map in seconds, targeting engineering teams who struggle with undocumented or out-of-sync APIs across services.
- In active development with open feedback loops from peer engineering teams.

KKA-DRAFT | Telkom University Student Directorate

April 2026 – June 2026

- Internal tool for Telkom University's student directorate enabling UKMs and student organizations to generate official Proposal Kegiatan and LPJ documents without the chaos of manual formatting.

- Delivered under a 3 month self-imposed sprint with full creative freedom and no predefined specification, with every design decision, feature, and UI flow owned end-to-end from scratch.
- Solved complex PDF rendering with dompdf including multi-page layouts, embedded tables, signatures, logos, and pixel-accurate formatting, replacing a previously manual process used across 50+ campus organizations.

HONORS & AWARDS

Finalist July 2026
OpenAI Build Week Hackathon 2026 | OpenAI

Finalist May 2026
Refactory Hackathon 2026 | Refactory

Finalist April 2026
Hackfest 2026 | Universitas Ciputra

CERTIFICATIONS

- Google Advanced Data Analytics Professional | Google
Issued: July 2026 | Credential ID: [NZH3N1FVRQVS](#)
- SQL (Advanced) | HackerRank
Issued: July 2026 | Credential ID: [4fb5ddcd42d8](#)
- Google AI Professional Certificate | Google
Issued: March 2026 | Credential ID: [WZVZJWMHBSNY](#)
- IBM AI Developer Professional Certificate | IBM
Issued: March 2026 | Credential ID: [QT3HO3K513XT](#)
- Microsoft Certified: Azure AI Fundamentals | Microsoft
Issued: September 2025 | Credential: [Credly](#)

LANGUAGES

- Indonesian: Native / Bilingual Proficiency.
- English: Professional Working Proficiency.